

# CELLAIR Mesoscope

CELL Analysis with Intelligent Recognition

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## What We Observe

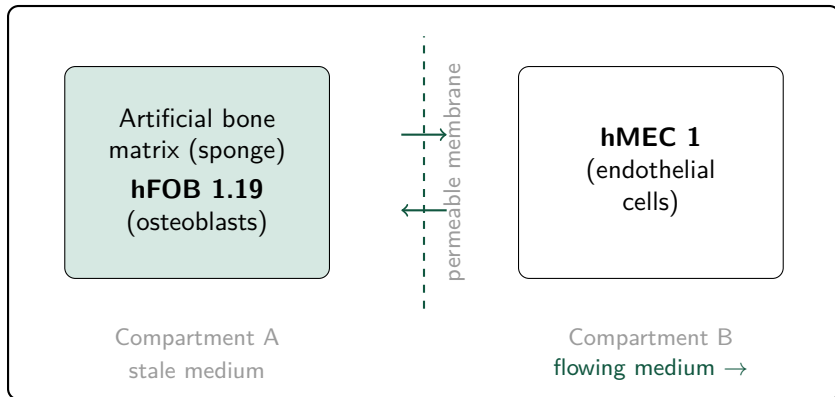
### Calcium production through cell–cell interaction

- ▶ **hFOB 1.19** (human fetal osteoblasts) and **hMEC 1** (human microvascular endothelial cells)
- ▶ Osteoblasts produce **hydroxyapatite** ( $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ ) — the mineral component of bone
- ▶ Key question: How do endothelial cells influence and respond to osteoblast mineralization?

Relevant for bone tissue engineering, implant osseointegration, and regenerative medicine.

## Technical Approach — Simulating the Natural Environment

Bioreactor Chip — 37 °C incubator



Flow rate: TBD (depends on chip geometry and shear stress tolerance)

## Current Status — What We Have

### **General experiment structure established.**

Bringing specimen and observation unit in place for early stage evaluation.

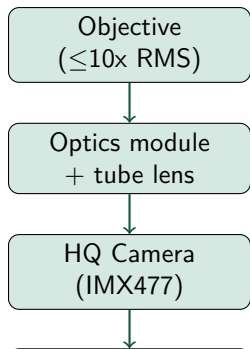
| <b>Component</b>                | <b>Status</b> |
|---------------------------------|---------------|
| Bioreactor chip design          | Established   |
| Cell lines (hFOB 1.19, hMEC 1)  | Available     |
| Computing (RPi 5 + Hailo-8 NPU) | Operational   |
| HQ Camera (IMX477, C-mount)     | Acquired      |
| OpenFlexure server (v3)         | Running       |

Currently verifying imaging quality on conventional microscope before transitioning to OpenFlexure platform.

# OpenFlexure Microscope — Hardware Platform

## Open-source, 3D-printed motorised microscope

Cambridge, CERN-OHL license



## Pending hardware:

| Part                   | Status   |
|------------------------|----------|
| Sangaboard v5          | On order |
| Stepper motors (3x)    | On order |
| Bag of Bits (hardware) | On order |
| Illumination kit       | On order |
| Tube lens (125–150 mm) | On order |
| Custom optics STL      | In dev.  |

All parts ordered via Taulab / distributor.  
Custom optics module compiled from OpenSCAD source.

## Why Custom Optics — The Sensor Utilization Problem

Standard OpenFlexure tube lens (50 mm) was designed for Pi Camera v2 (4.6 mm diagonal).

Our HQ Camera IMX477 has a **9.79 mm diagonal** — **2.1× larger**.

| Tube lens    | M     | EFN     | Sensor used |
|--------------|-------|---------|-------------|
| 50 mm (std.) | 0.261 | 37.5 mm | ~25%        |
| 125 mm       | 0.469 | 20.9 mm | ~85%        |
| 150 mm       | 0.515 | 19.0 mm | ~95%        |

Calculation:

$$M = \frac{f_t}{f_t + |p|}$$

$$\text{Area ratio} = \left( \frac{4.6}{9.79} \right)^2 \approx 22\%$$

$p = -141.5$  mm (back focal length)

EFN = effective field number

**Solution:** Replace 50 mm tube lens with **125–150 mm achromatic doublet** (same 12.7 mm diameter). Requires custom optics module STL.

## Prospect #1 — Enhancing Data Quality

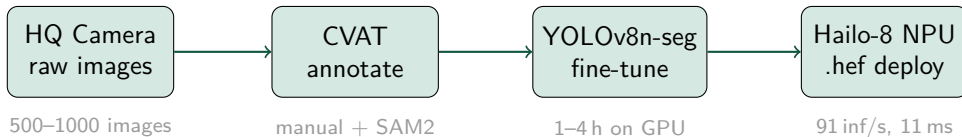
### Equipping experimental base setup with high-quality compounds

| Upgrade                       | Purpose                        | Impact                              |
|-------------------------------|--------------------------------|-------------------------------------|
| Custom optics module          | Match tube lens to IMX477      | Full sensor utilization             |
| $\leq 10\times$ RMS objective | Cell-level observation         | $\sim 1.3\mu\text{m}$ at $10\times$ |
| Flat-field correction         | Even illumination              | Consistent intensity                |
| Motorised stage               | Sub- $\mu\text{m}$ positioning | Automated time-lapse                |

**Result:** Higher quality raw data  $\rightarrow$  better annotation  $\rightarrow$  better model performance

## Prospect #2 — Automated Event Recognition

### Event-sensitive cell–cell interaction recognition via NPU inference



#### Target classes:

- ▶ Osteoblasts
- ▶ Endothelial cells
- ▶ Mineralization zones
- ▶ Membrane boundary

#### Annotation effort:

| Method        | Time/image |
|---------------|------------|
| Manual        | 5–10 min   |
| SAM2-assisted | 1–3 min    |

500 images  $\approx$  8–42 h  
1000 images  $\approx$  17–83 h

## Data Storage — Raw Image Estimates

**IMX477 sensor:**  $4056 \times 3040$  pixels, 12.3 MP

| Format      | /image  | 500     | 1000    |
|-------------|---------|---------|---------|
| 12-bit RAW  | 18.5 MB | 9.2 GB  | 18.5 GB |
| 16-bit TIFF | 24.7 MB | 12.3 GB | 24.7 GB |
| JPEG (HQ)   | 8 MB    | 4.0 GB  | 8.0 GB  |

**Time-lapse (48 h):**

| Interval | Frames | RAW   |
|----------|--------|-------|
| 1 min    | 2880   | 53 GB |
| 5 min    | 576    | 11 GB |
| 10 min   | 288    | 5 GB  |

### Recommendation

Attach **USB 3.0 SSD (256 GB+)** to RPi 5 for raw data collection. Back up to NAS after each session. SD card (32 GB) is *not* sufficient for RAW time-lapse series.

## Goal #1 — Biological Insight

### **Gaining deeper insight into cell–cell interaction dynamics during hydroxyapatite production**

- ▶ How does proximity to active osteoblasts affect endothelial cell morphology and migration?
- ▶ Does the rate of calcium deposition change in the presence of endothelial signaling?
- ▶ Can we identify temporal patterns — do interactions follow predictable phases?
- ▶ What morphological markers precede mineralization events?

### **Measurement approach:**

- ▶ Automated time-lapse imaging (hours to days)
- ▶ AI-assisted segmentation of cell populations and mineralization zones
- ▶ Quantitative tracking: cell count, area, migration velocity, mineralization density

## Goal #2 — Autonomous Reproducibility

### Building a stable, reproducible experimental platform

| Requirement              | Solution                                      |
|--------------------------|-----------------------------------------------|
| Hardware reproducibility | Open-source 3D-printed microscope (CERN-OHL)  |
| Software reproducibility | Ansible-automated deployment (<10 min)        |
| Protocol reproducibility | Scripted sessions via OpenFlexure REST API    |
| Analysis reproducibility | Compiled .hef model — deterministic inference |

### What this means

Any lab with a 3D printer and ~300 EUR can replicate the entire setup.  
Software stack is version-controlled. Imaging protocols are code.  
AI models are compiled binaries — identical results on any Hailo-8 NPU.

## Timeline

| Phase              | Deliverable                             | Status      |
|--------------------|-----------------------------------------|-------------|
| Experiment design  | Bioreactor chip, cell lines, protocol   | Established |
| Hardware base      | RPi 5, AI HAT, HQ Camera, OFM server    | Operational |
| Optics + stage     | Custom optics module, motorised stage   | In progress |
| Early evaluation   | First images on conventional microscope | Current     |
| Dataset collection | 500–1000 annotated time-lapse images    | Pending     |
| Model training     | YOLOv8n-seg fine-tuned on cell data     | Pending     |
| Deployment         | Real-time inference on Hailo-8 NPU      | Pending     |

## Open Questions

- ▶ **Flow rate:** Optimal flow rate ( $\mu\text{L}/\text{min}$ ) for Compartment B?  
Depends on chip geometry, pump, and shear stress tolerance of hMEC 1.
- ▶ **Objective:** Exact magnification of scraped objective to be verified —  
determines tube lens choice (125 mm vs. 150 mm).
- ▶ **Capture interval:** Optimal time-lapse interval for mineralization dynamics  
(5 min? 10 min?)
- ▶ **Training data:** Is a single experiment run sufficient for 500–1000 diverse images,  
or are multiple runs needed?

## Glossary

| Term           | Definition                                                          |
|----------------|---------------------------------------------------------------------|
| hFOB 1.19      | Human fetal osteoblast cell line                                    |
| hMEC 1         | Human microvascular endothelial cell line                           |
| Hydroxyapatite | $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ — primary bone mineral |
| OpenFlexure    | Open-source 3D-printed microscope (Cambridge)                       |
| RMS            | Royal Microscopical Society — objective thread standard             |
| Tube lens      | Achromat between objective and sensor — sets magnification          |
| YOLOv8n-seg    | YOLO v8 nano segmentation — lightweight edge model                  |
| Hailo HEF      | Compiled neural network format for Hailo NPU                        |
| NPU            | Neural Processing Unit (here: Hailo-8, 27 TOPS)                     |
| CVAT           | Computer Vision Annotation Tool (open source)                       |
| SAM2           | Segment Anything Model 2 — automatic pre-segmentation               |
| C-mount        | Camera mount standard, 17.526 mm FFD                                |

Thank you



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[openflexure.org](https://openflexure.org) · YOLOv8 · CVAT